

PAPER • OPEN ACCESS

Research on multi-sound source localization performance based on leaf-shaped microphone array

To cite this article: Ningning Wang *et al* 2023 *J. Phys.: Conf. Ser.* **2479** 012026

View the [article online](#) for updates and enhancements.

You may also like

- [Research on acoustic localization method for multiple leakage sources of water-cooling wall in furnace](#)
Qian Kong, Jiajia Yang, Yuechao Liu et al.
- [A high-precision DOA estimator for low-frequency signals using ultra-dense small-aperture microphone arrays](#)
Kunkun Zhang, Haonan Tong, Jun Tao et al.
- [Efficient energy-based orthogonal matching pursuit algorithm for multiple sound source localization with unknown source count](#)
Rongjiang Tang, Yingxiang Zuo, Weiya Liu et al.



ECS The Electrochemical Society
Advancing solid state & electrochemical science & technology

ECS UNITED

247th ECS Meeting
Montréal, Canada
May 18-22, 2025
Palais des Congrès de Montréal

Unite with the ECS Community

**Register to
save \$\$
before
May 17**

Research on multi-sound source localization performance based on leaf-shaped microphone array

Ningning Wang ^{a,b}, Yonghe Wei ^{*,a}, and Zonglin Li ^a

^a School of Mechanical Engineering, Shenyang Ligong University, Shenyang, 110159;

^b Kunshan Liaoyuan Automation Equipment Co., Ltd., Kunshan

* Corresponding author: yonghewe@sina.com

Abstract. In the research of multi-sound source localization, the geometric rules for the spatial arrangement of microphone arrays play a decisive role. When faced with multiple sound sources with close positions, traditional typical arrays are difficult to accurately locate the desired signals. Therefore, this paper proposes a leaf-shaped microphone array based on the theory of bionic sunflower spiral arrangement. When the incident signals are multiple sound sources with close positions, combined with the delay-sum beamforming algorithm, the performance of the leaf-shaped array is analyzed from the perspective of positioning accuracy and resolution. Compared with the uniform circular array and Arcondoulis spiral array, the results show that the leaf-shaped array has better resolution and anti-interference ability in the case of multiple sound sources with close positions.

1. INTRODUCTION

The microphone array combined with the beamforming algorithm can realize the functions of sound source identification, localization, enhancement, and so on, which provides strong support for the further processing of the sound signal to finally realize the equipment detection and fault diagnosis [1-3]. With the wide application of microphone arrays in various fields, the structure design of microphone arrays has also become a new research direction.

In the structure of microphone arrays, spiral arrays have been widely studied by researchers because of their advantages of non-uniform distribution between arrays which can avoid aliasing. In 2019, Ning Peipei [4] et al. conducted a simulation study about the Arcondoulis spiral array based on summarizing the shortcomings of other formations and concluded that it could obtain optimal results in the case of multi-source interference and low-frequency signals. However, the three sound sources in this simulation experiment are far away from each other, and the performance of the array in the presence of close sound sources is not analyzed. There are a large number of similar interference sound sources in the actual engineering environment. It was proposed that the design of a hybrid microphone array combined the portability and convenience benefits of a smaller spiral array design with the performance characteristics of a large array. Four improved spiral microphone arrays are proposed [5]. The array is constructed by placing the microphone on the axis of the point where the axis of the helical geometry is intercepted. However, each hybrid helix proposed in the paper has relatively high performance only for a specific environment. Some researchers proposed a general microphone array that can be selected from all available Pareto optimal designs with only one parameter [6]. But only the analysis of the beam width and the maximum side lobe level of single source beamforming were done. No multi-source localization studies were performed.



Some scholars optimize the array arrangement of a certain number of microphones in a specific environment through an optimization algorithm and use a digital optimization model to find the optimal solution. For example, Le [7] used a simulated annealing algorithm to find an irregular microphone array. Le et al. [8] used a genetic algorithm to optimize the geometric structure of the microphone array. Liu Haitao [9] and others used a POS optimization algorithm to optimize the microphone array structure. In the optimization model, a reasonable fitness evaluation function is established which can comprehensively consider the positioning accuracy and robustness of the microphone array. The performance of the microphone array obtained by using the optimization algorithm is improved, but it also brings a lot of computation, so it is difficult to be widely used.

Therefore, it is necessary to study a microphone array structure with high performance and general applicability which is no need for complex algorithm optimization. Lv Yushan [10-11] et al. designed the abrasive arrangement on the surface of the grinding wheel based on the sunflower spiral arrangement principle and used the optimal geometric complementary principle of the sunflower spiral arrangement to achieve a good grinding effect of the grinding wheel. Combining the ideas, this paper proposes a new formation based on the theory of bionic leaf arrangement. The microphone array is arranged according to the natural laws that sunflower seeds follow. The performance of the leaf-shaped microphone array is analyzed from the perspectives of resolution and positioning accuracy in the presence of multiple close sound sources. Compared with the uniform circular array and the spiral array, the results show that under the same conditions, the leaf-shaped array still has better resolution and positioning accuracy when the desired signal has multiple nearby interfering sound sources.

2. MICROPHONE ARRAY

2.1 Sunflower spiral array

The sunflower spiral is the distribution pattern of sunflower seeds. The growth sequence of sunflower seeds has a certain order, and it grows according to the golden angle. The angle between the latter seed and the former seed is called the golden angle. The golden angle is about 137.508° [12]. Many experts and scholars have shown that the spiral arrangement of sunflowers satisfies the model of Vogel [13-14].

$$r = R\sqrt{\frac{m}{M}} \quad m = 1, 2, \dots, M \quad (1)$$

$$\theta = 2\pi m \frac{(1 + \sqrt{V})}{2} \quad V = 5 \quad (2)$$

The above formula is the polar coordinate form of the sunflower spiral array. Where, R is the radius of the array aperture and M is the number of array elements. According to the growth order of the seeds, the first 16 points of the seed growth are selected as the positions of the array elements to form the leaf-shaped array proposed in this paper. The arrangement of the leaf-shaped array is shown in Figure 1.

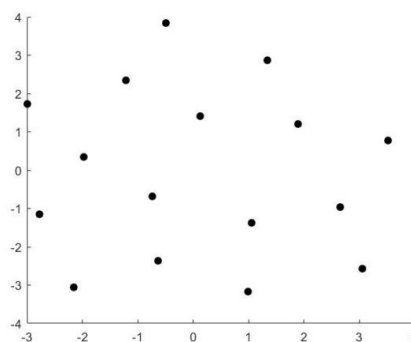


Figure 1 Leaf-shaped array

3. BEAMFORMING PRINCIPLE BASED ON SHAPED ARRAY

3.1 Beamforming based on the leaf-shaped array

Assuming that there are N signals in the space. The incident signals are denoted by $s_i(t)$, where $i=1, 2, \dots, N$, and the coordinates of the sound source are (x_i, y_i, z_i) . The number of array elements is M , and the Cartesian coordinates of the leaf-shaped array are $(x_j, 0, z_j)$, where $j = (1, 2, \dots, M)$. The Cartesian coordinates of the leaf-shaped array in the x - z plane are:

$$\begin{cases} x_j = r \cos(\theta) \\ z_j = r \sin(\theta) \end{cases} \quad (3)$$

Substituting Equation (1) and Equation (2) into Equation (4) to obtain the rectangular coordinate model of the leaf-shaped array. Then the distance from the sound source to the array is obtained by the Euclidean distance formula:

$$r_j = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} \quad (4)$$

When the frequency is ω , the direction vector of the incident signal $s_i(t)$ is:

$$a_j = \frac{r_i}{r_{ij}} e^{-j\omega \frac{(r_{ij} - r_i)}{c}} \quad (5)$$

Where r_i is the distance from the i th sound source to the coordinate origin, and r_{ij} is the distance from the i th sound source to the j th microphone. Where $(r_{ij} - r_i)$ is the sound path difference of the signal received by each array element, and c is the wave speed. Then the array flow pattern vector can be obtained, where $A = (a_1, a_2, \dots, a_M)$. The output signal based on the snapshot data model can be expressed as:

$$y(n) = W^H X(n) \quad n = 1, 2, 3, \dots, N \quad (6)$$

Where $W = [a_1 \ a_2 \ \dots \ a_M]$, and the output power of the array can be expressed as:

$$P = E[|y(n)|^2] = W^H R_{XX} W \quad (7)$$

Where R_{XX} is the covariance matrix of the received signal.

4. EXPERIMENTAL VERIFICATION

4.1 Experiment 1: Comparison of positioning accuracy in the presence of interference

This section mainly analyzes the single source positioning accuracy of each microphone array when there are close interference sources in the desired signal. Three sinusoidal signals with frequencies of 1000HZ, 300HZ, and 100HZ respectively added with Gaussian white noise were selected as the incident signals of the array.

(1) Sound source position is $(2, 4, 2)$, and the signal frequency is 1000HZ

A sinusoidal signal with a frequency of 1000HZ and the position of $(2, 4, 2)$ is taken as the desired signal. Two signals with frequencies of 300HZ and 100HZ are used as interference signals which are placed at two positions $(2.5, 4, 2.5)$ and $(1, 4, 1)$ respectively. Using the delay-sum beamforming algorithm to form a beam in the direction of the desired signal, and scanning on the $(x, 4, z)$ plane. The

scan step size is set to 0.1, and the scan ranges of the x-axis and z-axis are both $(-3 < x < 3)$. The single sound source localization results of each formation are shown in Figure 2.

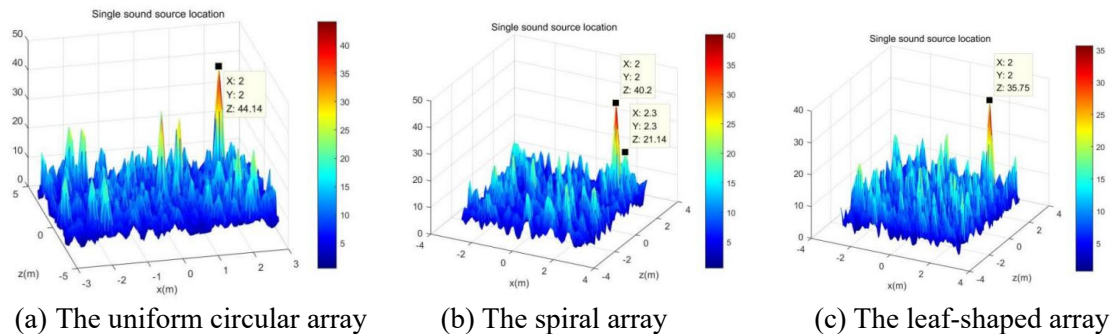


Figure 2 The result of single sound source localization at (2, 2)

It can be seen from Figure 2 that the single sound source localization based on the uniform circular array fails when the signal frequency is high and the signal with near interfering sound sources. The result of the spiral array forms a beam at (2, 2), but at (2.3, 2.3) a higher side lobe is formed at the position. The leaf-shaped array successfully suppresses the interference and forms a beam in the desired direction (2, 2).

(2) Sound source position is (2.5, 2.5), and the signal frequency is 300HZ

The signal with a frequency of 300HZ and the position of (2.5, 2.5) is used as the desired signal. The other two signals are regarded as interference signals. Other conditions are the same as described in (1). The single sound source localization results are shown in Figure 3:

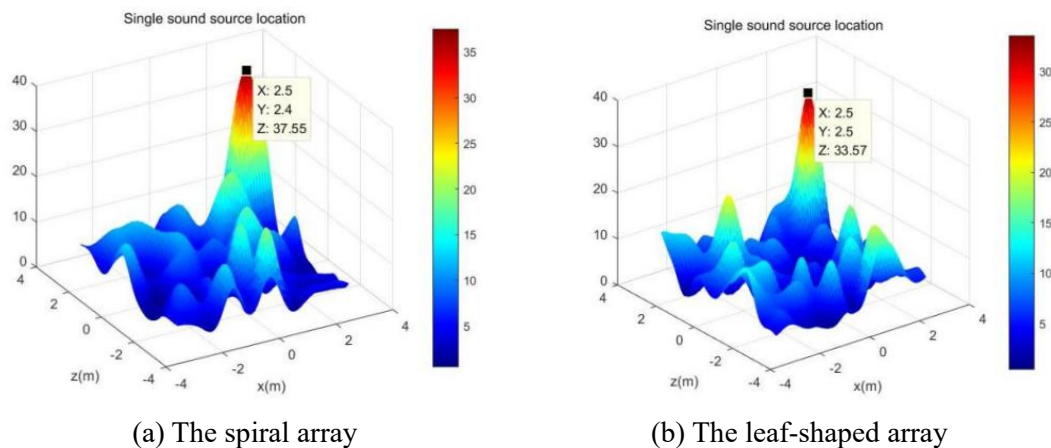
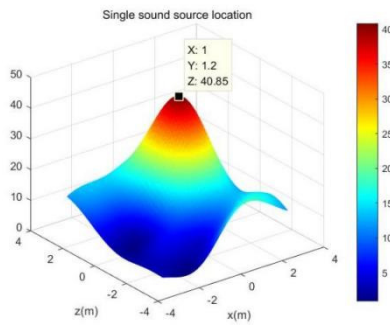


Figure 3 The result of single sound source localization at (2.5, 2.5)

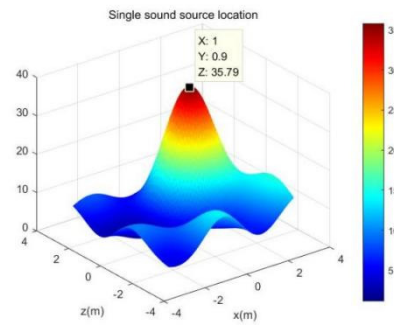
It can be seen from Figure 3 that when the signal frequency is 300HZ, the spiral array can form a beam in the desired direction, but there is an error in the positioning result. The leaf-shaped array can still accurately form a beam in the desired direction, while suppression is formed in the direction of interference.

(3) Sound source position is (1,1), and the signal frequency is 100HZ

The signal with the frequency of 100HZ and the position at (1, 1) is used as the desired signal, and the two signals at positions (2, 2) and (2.5, 2.5) as the interference. The mixed signals are received by the spiral array and the leaf-shaped array respectively. Other conditions are the same as described in (1). The single sound source localization results are shown in Figure 4.



(a) The spiral array



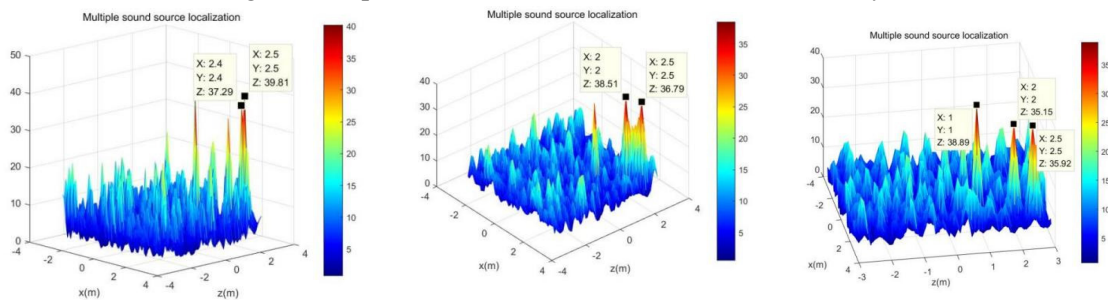
(b) The leaf-shaped array

Figure 4 The result of single sound source localization at (1, 1)

As can be seen from Figure 4 that the single sound source localization results based on the spiral array and the leaf-shaped array have localization errors, but the positioning error of the latter is smaller than that of the former.

4.2 Experiment 2: Multi-source localization comparison resolution

In this section, the beamforming algorithm is used to locate three close sound sources to verify the resolution of different formations. The three sound sources are sinusoidal signals with frequencies of 1000HZ, 300HZ, and 100HZ respectively. The spatial position coordinates of each sound source are (2, 4, 2), (2.5, 4, 2.5), and (1, 4, 1). The three sound sources are respectively added with Gaussian white noise to form a mixed signal. The positioning results of each microphone array are as follows.



(a) The uniform circular array (b) The spiral array (c) The leaf-shaped array

Figure 5 The results of multi-sound source localization

As shown in Figure 5 the uniform circular array can locate three desired signals, but there is also an obvious false alarm phenomenon at the position (2.4, 2.4). The result of the spiral array produces large interference side lobes between (2, 2) and (2.5, 2.5) which can improve the false alarm probability of target detection. The leaf-shaped array can accurately locate the positions of the three sound sources, and the side lobe interference is very small.

5. Conclusion

In the past, it is difficult to accurately locate the desired signals when facing multiple similar sound sources combined with the traditional beamforming algorithm. Changing the algorithm to improve the positioning accuracy will also increase the amount of calculation. Therefore, according to the theory of bionic sunflower spiral arrangement, this paper designs a leaf-shaped array arranged according to the growth pattern of sunflower spiral seeds. The first 16 growth positions of sunflower seeds are selected as the arrangement positions of the 16 microphones in the leaf-shaped array. Combined with the delay-sum beamforming algorithm, the performance of the leaf-shaped array was analyzed from two angles, and compared with the uniform circular array and spiral array which have been widely used. The simulation results of Experiment 1 and Experiment 2 show when there are close interference sources in the signals, the leaf-shaped array has better positioning accuracy and better interference suppression

ability than other arrays. When there are multiple close sound sources, the leaf-shaped array has better resolution than the other two arrays.

REFERENCES

- [1] Marques I, Sousa J, Sá B, et al. Microphone Array for Speaker Localization and Identification in Shared Autonomous Vehicles [J]. *Electronics*, 2022, 11(5): 766.
- [2] ZHOU J J, HAO X, CHEN D B, et al. Research of moving noise source recognition based on microphone array [J]. *Acta Aerodynamica Sinica*, 2022, 40: 1–10 (in Chinese). doi: 10.7638/kqdlxxb-2020.0127
- [3] Zhao Y, Chu Z, Li L. Performance Enhancement of Functional Delay and Sum Beamforming for Spherical Microphone Arrays [J]. *Electronics*, 2022, 11(7): 1132.
- [4] Ning Peipei. Research on mechanical fault diagnosis based on microphone array [D]. University of Electronic Science and Technology of China, 2019.
- [5] Aldeman M R. A hybrid spiral microphone array design for performance and portability [J]. *Applied Acoustics*, 2020, 170: 107512.
- [6] Sarraji E. A generic approach to synthesize optimal array microphone arrangements[C]//*Proceedings of the 6th Berlin beamforming conference*. 2016, 29.
- [7] Le Son P. Irregular microphone array design for broadband beamforming [J]. *Signal Processing*, 2022, 193: 108431.
- [8] Le Courtois F, Thomas JH, Poisson F, et al. Genetic optimization of plane array geometry for beamforming. Application to source localization in a high-speed train [J]. *Journal of Sound and Vibration*, 2016, 371: 78 -93.
- [9] Liu H, Kirubarajan T, Xiao Q. Arbitrary microphone array optimization method based on TDOA for specific localization scenarios [J]. *Sensors*, 2019, 19(19): 4326.
- [10] Lv Yushan, Wang Jun, Zhao Chengyi, He Yan. An End-grinding Wheel with the Phyllotactic Pattern of Abrasive Grain Clusters and Its Mill-grinding Experiments [J]. *ACTA ARMAMENTARII*, 2013, 34(12): 1569-1574.
- [11] Tao Siyuan, Lv Yushan, Wang Yuchen, Song Zheng. Simulation Study on Grinding Microstructured Surface by the Grinding Wheels with Abrasive Phyllotactic Pattern[J]. *Modular Machine Tool & Automatic Manufacturing Technique*, 2020(09): 130-133+138. DOI:10.13462/j.cnki.mmtamt.2020.09.029.
- [12] Yu Haiyue. Research on the mechanism of the grinding wheel with an abrasive phyllotactic pattern [D]. Northeastern University, 2017.
- [13] Vogel H. A better way to construct the sunflower head [J]. *Mathematical biosciences*, 1979, 44(3-4): 179-189.
- [14] Puhle C, Hollands S. On the sunflower spiral: acoustical holography results[C]//*INTER-NOISE and NOISE-CON Congress and Conference Proceedings*. Institute of Noise Control Engineering, 2016, 253(3): 5715-5726.